Project

Title: Multi-Drone Disaster Assessment and Rescue Planning

Background

Following a major flood, a disaster management authority deploys autonomous drones to assess affected regions and identify people requiring immediate assistance. The affected area is represented as a grid where each cell may contain victims, obstacles, or safe zones.

Each drone has limited battery capacity, communication range, and carrying capability. Due to damaged infrastructure and uncertain conditions, the authority requires an efficient algorithmic solution to maximize rescue effectiveness.

Design and implement an algorithm that coordinates multiple drones to survey the disaster area and prioritize rescue operations.

The disaster area is represented by N×N grid where:

- `0` = Accessible area
- `X` = Obstacle / No-fly zone
- `V` = Victim location
- `B` = Drone base station

Each drone:

- Starts from the base station.
- Has a fixed battery capacity allowing a limited number of movements.
- Must return to the base before battery depletion.
- Can communicate only within a specified communication range.

Input Parameters

Your algorithm should accept:

1. Grid size N
2. Number of drones D
3. Battery capacity of each drone
4. Communication range
5. Disaster area map
6. Victim priority level:
    - High Priority (3)
    - Medium Priority (2)
    - Low Priority (1)

Objectives

Design an algorithm that:

1. Maximizes the number of victims identified.
2. Prioritizes high-priority victims.
3. Maximizes total area coverage.
4. Ensures all drones safely return to the base station.
5. Minimizes unnecessary movements and battery consumption.

Constraints

- Grid size: $20 \le N \le 100$
- Number of drones: $2 \le D \le 20$
- Obstacles may block movement.
- Victim locations are initially unknown to drones and are discovered during exploration.
- A drone that runs out of battery before returning is considered lost.

Tasks

Part A: Problem Analysis

Identify:

- Inputs
- Outputs
- Constraints
- Assumptions

Part B: Algorithm Design

Develop an algorithm that:

- Assigns exploration regions to drones.
- Determines movement decisions.
- Handles battery limitations.
- Updates rescue priorities dynamically.

Provide:

- Flowchart
- Pseudocode

Part C: Complexity Analysis

Determine:

- Time Complexity
- Space Complexity

Part D: Implementation

Implement your proposed solution using any programming language.

Part E: Experimental Evaluation

Test your algorithm on at least three different disaster scenarios:

Scenari	Grid Size	Drone	Victim	Obstacle
1	20×20	3	15	20
2	50×50	5	50	100
3	100×100	10	150	500

Compare:

- Victims found
- Coverage percentage
- Battery utilization
- Execution time

Part F: Discussion

Discuss:

- Strengths of your solution
- Limitations
- Possible improvements
- A simplified justification of how this problem aligns with complex engineering problem.